Name: Maher Atef Maher

Lab : 4

1- Display all students' full names only using aggregation.

```
FacultySystemV2> db.students.aggregate([
| {
|  $project:{
|    fullName:{ $concat:["$firstName"," ","$lastName"] }
|  }
| }
| ])
[
  { _id: ObjectId('69aabd3317ec58078b7c2907'), fullName: 'Ahmed Ali' },
  { _id: ObjectId('69aabd3317ec58078b7c2908'), fullName: 'Sara Hassan' },
  { _id: ObjectId('69aabd3317ec58078b7c2909'), fullName: 'Omar Mahmoud' }
]
```

2- Display students who are not fired.

```
FacultySystemV2> db.students.aggregate([
| {
|  $match:{ isFired:false }
| }
| ])
[
  {
    _id: ObjectId('69aabd3317ec58078b7c2907'),
    firstName: 'Ahmed',
    lastName: 'Ali',
    isFired: false,
    facultyId: 1,
    courses: [ { courseId: 101, grade: 90 }, { courseId: 102, grade: 80 } ]
  },
  {
    _id: ObjectId('69aabd3317ec58078b7c2908'),
    firstName: 'Sara',
    lastName: 'Hassan',
    isFired: false,
    facultyId: 2,
    courses: [ { courseId: 101, grade: 85 }, { courseId: 103, grade: 95 } ]
  }
]
```

3- Display students sorted by lastName ascending.

```
FacultySystemV2> db.students.aggregate([
| {
|  $sort:{ lastName:1 }
| }
| ])
[
  {
    _id: ObjectId('69aabd3317ec58078b7c2907'),
    firstName: 'Ahmed',
    lastName: 'Ali',
    isFired: false,
    facultyId: 1,
    courses: [ { courseId: 101, grade: 90 }, { courseId: 102, grade: 80 } ]
  },
  {
    _id: ObjectId('69aabd3317ec58078b7c2908'),
    firstName: 'Sara',
    lastName: 'Hassan',
    isFired: false,
    facultyId: 2,
    courses: [ { courseId: 101, grade: 85 }, { courseId: 103, grade: 95 } ]
  },
  {
    _id: ObjectId('69aabd3317ec58078b7c2909'),
    firstName: 'Omar',
    lastName: 'Mahmoud',
    isFired: true,
    facultyId: 1,
    courses: [ { courseId: 102, grade: 70 }, { courseId: 104, grade: 60 } ]
  }
]
```

4- Show only first 2 students.

```
FacultySystemV2> db.students.aggregate([
| {
|  $limit:2
| }
| ])
[
  {
    _id: ObjectId('69aabd3317ec58078b7c2907'),
    firstName: 'Ahmed',
    lastName: 'Ali',
    isFired: false,
    facultyId: 1,
    courses: [ { courseId: 101, grade: 90 }, { courseId: 102, grade: 80 } ]
  },
  {
    _id: ObjectId('69aabd3317ec58078b7c2908'),
    firstName: 'Sara',
    lastName: 'Hassan',
    isFired: false,
    facultyId: 2,
    courses: [ { courseId: 101, grade: 85 }, { courseId: 103, grade: 95 } ]
  }
]
```

5- Display each student:

=> full name

=> number of courses he takes

```
FacultySystemV2> db.students.aggregate([
| {
|  $project:{
|   fullName:{ $concat:["$firstName"," ","$lastName"] },
|   coursesCount:{ $size:"$courses" }
|  }
| }
| ])
[
  {
    _id: ObjectId('69aabd3317ec58078b7c2907'),
    fullName: 'Ahmed Ali',
    coursesCount: 2
  },
  {
    _id: ObjectId('69aabd3317ec58078b7c2908'),
    fullName: 'Sara Hassan',
    coursesCount: 2
  },
  {
    _id: ObjectId('69aabd3317ec58078b7c2909'),
    fullName: 'Omar Mahmoud',
    coursesCount: 2
  }
]
```

6- Display each student's:

=> full name

=> average grade in all courses

```
FacultySystemV2> db.students.aggregate([
| {
|  $project:{
|   fullName:{ $concat:["$firstName"," ","$lastName"] },
|   avgGrade:{ $avg:"$courses.grade" }
|  }
| }
| ])
[
  {
    _id: ObjectId('69aabd3317ec58078b7c2907'),
    fullName: 'Ahmed Ali',
    avgGrade: 85
  },
  {
    _id: ObjectId('69aabd3317ec58078b7c2908'),
    fullName: 'Sara Hassan',
    avgGrade: 90
  },
  {
    _id: ObjectId('69aabd3317ec58078b7c2909'),
    fullName: 'Omar Mahmoud',
    avgGrade: 65
  }
]
```

7- Display only students whose average grade > 80.

```
FacultySystemV2> db.students.aggregate([
| {
|  $project:{
|   fullName:{ $concat:["$firstName"," ","$lastName"] },
|   avgGrade:{ $avg:"$courses.grade" }
|  }
| },
| {
|  $match:{ avgGrade:{ $gt:80 } }
| }
| ])
[
  {
    _id: ObjectId('69aabd3317ec58078b7c2907'),
    fullName: 'Ahmed Ali',
    avgGrade: 85
  },
  {
    _id: ObjectId('69aabd3317ec58078b7c2908'),
    fullName: 'Sara Hassan',
    avgGrade: 90
  }
]
```

8- Display each course taken by students as separate documents (one course per row).

```
FacultySystemV2> db.students.aggregate([
| {
|  $unwind:"$courses"
| }
| ])
[
  {
    _id: ObjectId('69aabd3317ec58078b7c2907'),
    firstName: 'Ahmed',
    lastName: 'Ali',
    isFired: false,
    facultyId: 1,
    courses: { courseId: 101, grade: 90 }
  },
  {
    _id: ObjectId('69aabd3317ec58078b7c2907'),
    firstName: 'Ahmed',
    lastName: 'Ali',
    isFired: false,
    facultyId: 1,
    courses: { courseId: 102, grade: 80 }
  },
  {
    _id: ObjectId('69aabd3317ec58078b7c2908'),
    firstName: 'Sara',
    lastName: 'Hassan',
    isFired: false,
    facultyId: 2,
    courses: { courseId: 101, grade: 85 }
  },
  {
    _id: ObjectId('69aabd3317ec58078b7c2908'),
    firstName: 'Sara',
    lastName: 'Hassan',
    isFired: false,
    facultyId: 2,
    courses: { courseId: 103, grade: 95 }
```

9- Count total number of students.

```
FacultySystemV2> db.students.aggregate([
| {
|  $count:"totalStudents"
| }
| ])
[ { totalStudents: 3 } ]
```

10- Calculate total number of courses registrations across all students.

```
FacultySystemV2> db.students.aggregate([
| {
|  $unwind:"$courses"
| },
| {
|  $count:"totalRegistrations"
| }
| ])
[ { totalRegistrations: 6 } ]
```

11- Find highest grade achieved in the system.

```
FacultySystemV2> db.students.aggregate([
| {$unwind:"$courses"},
| {
|  $group:{
|   _id:null,
|   maxGrade:{ $max:"$courses.grade" }
|  }
| }
| ])
[ { _id: null, maxGrade: 95 } ]
```

12- Find lowest grade achieved in the system.

```
FacultySystemV2> db.students.aggregate([
| {$unwind:"$courses"},
| {
|  $group:{
|   _id:null,
|   minGrade:{ $min:"$courses.grade" }
|  }
| }
| ])
[ { _id: null, minGrade: 60 } ]
```

13- Calculate average grade per courseId across all students.

```
FacultySystemV2> db.students.aggregate([
| {$unwind:"$courses"},
| {
|  $group:{
|   _id:"$courses.courseId",
|   avgGrade:{ $avg:"$courses.grade" }
|  }
| }
| ])
[
  { _id: 103, avgGrade: 95 },
  { _id: 104, avgGrade: 60 },
  { _id: 101, avgGrade: 87.5 },
  { _id: 102, avgGrade: 75 }
]
```

14- Group students by isFired and display count for each group.

```
FacultySystemV2> db.students.aggregate([
| {
|  $group:{
|   _id:"$isFired",
|   count:{ $sum:1 }
|  }
| }
| ])
[ { _id: true, count: 1 }, { _id: false, count: 2 } ]
```

15- Show how many students belong to each faculty (group by facultyId).

```
FacultySystemV2> db.students.aggregate([
| {
|  $group:{
|   _id:"$facultyId",
|   studentsCount:{ $sum:1 }
|  }
| }
```

16- For each student, display:

=> full name

=>faculty name

(using lookup)

```
FacultySystemV2> db.students.aggregate([
| {
|  $lookup:{
|   from:"faculty",
|   localField:"facultyId",
|   foreignField:"_id",
|   as:"faculty"
|  }
| },
| {$unwind:"$faculty"},
| {
|  $project:{
|   fullName:{ $concat:["$firstName"," ","$lastName"] },
|   facultyName:"$faculty.name"
|  }
| }
| ])
[
  {
    _id: ObjectId('69aabd3317ec58078b7c2907'),
    fullName: 'Ahmed Ali',
    facultyName: 'Engineering'
  },
  {
    _id: ObjectId('69aabd3317ec58078b7c2908'),
    fullName: 'Sara Hassan',
    facultyName: 'Commerce'
  },
  {
    _id: ObjectId('69aabd3317ec58078b7c2909'),
    fullName: 'Omar Mahmoud',
    facultyName: 'Engineering'
  }
]
```

17- Select one specific student by name and display:

=> his courses names instead of courseIds

```
FacultySystemV2> db.students.aggregate([
| {
|  $match:{ firstName:"Ahmed" }
| },
| {$unwind:"$courses"},
| {
|  $lookup:{
|   from:"courses",
|   localField:"courses.courseId",
|   foreignField:"_id",
|   as:"course"
|  }
| },
| {$unwind:"$course"},
| {
|  $project:{
|   courseName:"$course.name",
|   grade:"$courses.grade"
|  }
| }
| ])
[
  {
    _id: ObjectId('69aabd3317ec58078b7c2907'),
    courseName: 'Math',
    grade: 90
  },
  {
    _id: ObjectId('69aabd3317ec58078b7c2907'),
    courseName: 'Physics',
    grade: 80
  }
]
```

18-Display all students with:

=> full name

=> faculty name

=> list of course names

(3 collections join)

```
FacultySystemV2> db.students.aggregate([
  {
   $lookup:{
    from:"faculty",
    localField:"facultyId",
    foreignField:"_id",
    as:"faculty"
   }
  },
  {$unwind:"$faculty"},
  {$unwind:"$courses"},
  {
   $lookup:{
    from:"courses",
    localField:"courses.courseId",
    foreignField:"_id",
    as:"course"
   }
  },
  {$unwind:"$course"},
  {
   $group:{
    _id:"$_id",
    fullName:{ $first:{ $concat:["$firstName"," ","$lastName"] }},
    faculty:{ $first:"$faculty.name"},
    courses:{ $push:"$course.name"}
   }
  }
  ])

  {
    _id: ObjectId('69aabd3317ec58078b7c2907'),
    fullName: 'Ahmed Ali',
    faculty: 'Engineering',
    courses: [ 'Math', 'Physics' ]
  },
  {
    _id: ObjectId('69aabd3317ec58078b7c2909'),
    fullName: 'Omar Mahmoud',
    faculty: 'Engineering',
    courses: [ 'Physics', 'Database' ]
  },
  {
```

19- Show students who belong to Engineering faculty only.

```
FacultySystemV2> db.students.aggregate([
| {
|  $lookup:{
|   from:"faculty",
|   localField:"facultyId",
|   foreignField:"_id",
|   as:"faculty"
|  }
| },
| {$unwind:"$faculty"},
| {
|  $match:{ "faculty.name":"Engineering" }
| }
| ])
[
  {
    _id: ObjectId('69aabd3317ec58078b7c2907'),
    firstName: 'Ahmed',
    lastName: 'Ali',
    isFired: false,
    facultyId: 1,
    courses: [ { courseId: 101, grade: 90 }, { courseId: 102, grade: 80 } ],
    faculty: { _id: 1, name: 'Engineering', address: 'Cairo' }
  },
  {
    _id: ObjectId('69aabd3317ec58078b7c2909'),
    firstName: 'Omar',
    lastName: 'Mahmoud',
    isFired: true,
    facultyId: 1,
    courses: [ { courseId: 102, grade: 70 }, { courseId: 104, grade: 60 } ],
    faculty: { _id: 1, name: 'Engineering', address: 'Cairo' }
  }
```

20- Display courses that no students registered for.

```
FacultySystemV2> db.courses.aggregate([
| {
|  $lookup:{
|   from:"students",
|   localField:"_id",
|   foreignField:"courses.courseId",
|   as:"students"
|  }
| },
| {
|  $match:{ students:{ $size:0 } }
| }
| ])
```

21- Calculate sum of final marks for all courses.

```
FacultySystemV2> db.courses.aggregate([
| {
|  $group:{
|   _id:null,
|   total:{ $sum:"$finalMark" }
|  }
| }
| ])
[ { _id: null, total: 400 } ]
```

22- Calculate total grades per student.

```
FacultySystemV2> db.students.aggregate([
| {
|  $project:{
|   fullName:{ $concat:["$firstName"," ","$lastName"] },
|   totalGrades:{ $sum:"$courses.grade" }
|  }
| }
| ])
[
  {
    _id: ObjectId('69aabd3317ec58078b7c2907'),
    fullName: 'Ahmed Ali',
    totalGrades: 170
  },
  {
    _id: ObjectId('69aabd3317ec58078b7c2908'),
    fullName: 'Sara Hassan',
    totalGrades: 180
  },
  {
    _id: ObjectId('69aabd3317ec58078b7c2909'),
    fullName: 'Omar Mahmoud',
    totalGrades: 130
  }
]
```

23- Show top 3 students by average grade.

```
FacultySystemV2> db.students.aggregate([
| {
|  $project:{
|   fullName:{ $concat:["$firstName"," ","$lastName"] },
|   avg:{ $avg:"$courses.grade" }
|  }
| },
| {$sort:{ avg:-1 }},
| {$limit:3}
| ])
[
  {
    _id: ObjectId('69aabd3317ec58078b7c2908'),
    fullName: 'Sara Hassan',
    avg: 90
  },
  {
    _id: ObjectId('69aabd3317ec58078b7c2907'),
    fullName: 'Ahmed Ali',
    avg: 85
  },
  {
    _id: ObjectId('69aabd3317ec58078b7c2909'),
    fullName: 'Omar Mahmoud',
    avg: 65
  }
]
```

24- Show bottom 2 students by average grade.

```
FacultySystemV2> db.students.aggregate([
| {
|  $project:{
|   fullName:{ $concat:["$firstName"," ","$lastName"] },
|   avg:{ $avg:"$courses.grade" }
|  }
| },
| {$sort:{ avg:1 }},
| {$limit:2}
| ])
[
  {
    _id: ObjectId('69aabd3317ec58078b7c2909'),
    fullName: 'Omar Mahmoud',
    avg: 65
  },
  {
    _id: ObjectId('69aabd3317ec58078b7c2907'),
    fullName: 'Ahmed Ali',
    avg: 85
  }
]
```

25- For each faculty, calculate:

=> number of students

=> average grade of its students

```
FacultySystemV2> db.students.aggregate([
| {$unwind:"$courses"},
| {
|  $group:{
|   _id:"$facultyId",
|   studentsCount:{ $addToSet:"$_id" },
|   avgGrade:{ $avg:"$courses.grade" }
|  }
| },
| {
|  $project:{
|   studentsCount:{ $size:"$studentsCount" },
|   avgGrade:1
|  }
| }
| ])
[
  { _id: 1, avgGrade: 75, studentsCount: 2 },
  { _id: 2, avgGrade: 90, studentsCount: 1 }
]
```

26- Add a new field called status:

=> "Pass" if avg ≥ 60

=> "Fail" otherwise

```
FacultySystemV2> db.students.aggregate([
| {
|  $project:{
|   fullName:{ $concat:["$firstName"," ","$lastName"] },
|   avg:{ $avg:"$courses.grade" },
|   status:{
|    $cond:[
|     { $gte:[{ $avg:"$courses.grade" },60]},
|     "Pass",
|     "Fail"
|    ]
|   }
|  }
| }
| ])
[
  {
    _id: ObjectId('69aabd3317ec58078b7c2907'),
    fullName: 'Ahmed Ali',
    avg: 85,
    status: 'Pass'
  },
  {
    _id: ObjectId('69aabd3317ec58078b7c2908'),
    fullName: 'Sara Hassan',
    avg: 90,
    status: 'Pass'
  },
  {
    _id: ObjectId('69aabd3317ec58078b7c2909'),
    fullName: 'Omar Mahmoud',
    avg: 65,
    status: 'Pass'
  }
]
```

27- Remove sensitive fields (isFired, facultyId) from output.

```
FacultySystemV2> db.students.aggregate([
| {
|  $project:{
|   isFired:0,
|   facultyId:0
|  }
| }
| ])
[
  {
    _id: ObjectId('69aabd3317ec58078b7c2907'),
    firstName: 'Ahmed',
    lastName: 'Ali',
    courses: [ { courseId: 101, grade: 90 }, { courseId: 102, grade: 80 } ]
  },
  {
    _id: ObjectId('69aabd3317ec58078b7c2908'),
    firstName: 'Sara',
    lastName: 'Hassan',
    courses: [ { courseId: 101, grade: 85 }, { courseId: 103, grade: 95 } ]
  },
  {
    _id: ObjectId('69aabd3317ec58078b7c2909'),
    firstName: 'Omar',
    lastName: 'Mahmoud',
    courses: [ { courseId: 102, grade: 70 }, { courseId: 104, grade: 60 } ]
  }
]
```

28- Create a report that returns:

=> total students

=> total courses

=> total faculties

(in one query)

```
FacultySystemV2> db.students.aggregate([
| {
|  $facet:{
|   totalStudents:[{ $count:"count"}],
|   totalCourses:[{ $lookup:{from:"courses",pipeline:[{$count:"count"}],as:"courses"}}],
|   totalFaculties:[{ $lookup:{from:"faculty",pipeline:[{$count:"count"}],as:"faculties"}}]
|  }
| }
| ])
[
  {
    totalStudents: [ { count: 3 } ],
    totalCourses: [
      {
        _id: ObjectId('69aabd3317ec58078b7c2907'),
        firstName: 'Ahmed',
        lastName: 'Ali',
        isFired: false,
        facultyId: 1,
        courses: [ { count: 4 } ]
      },
      {
        _id: ObjectId('69aabd3317ec58078b7c2908'),
        firstName: 'Sara',
        lastName: 'Hassan',
        isFired: false,
        facultyId: 2,
        courses: [ { count: 4 } ]
      },
      {
        _id: ObjectId('69aabd3317ec58078b7c2909'),
        firstName: 'Omar',
        lastName: 'Mahmoud',
        isFired: true,
        facultyId: 1,
        courses: [ { count: 4 } ]
      }
    ],
    totalFaculties: [
      {
        _id: ObjectId('69aabd3317ec58078b7c2907'),
        firstName: 'Ahmed',
        lastName: 'Ali',
        isFired: false,
```